



O2) Controllability and observability

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

Controllability matrix U_c :

$$U_c = \begin{bmatrix} [B] & [AB] & [A^2B] & \dots & [A^{n-1}B] \end{bmatrix}$$

if matrix rank of U_c is n , the pair (A, B) is controllable

Controllable or Stabilizable $\Rightarrow$ There exists a feedback gain that stabilizes the control system.

Observability Matrix U_o :

$$U_o = \begin{bmatrix} [C] \\ [CA] \\ \vdots \\ [CA^{n-1}] \end{bmatrix}$$

if rank of U_o is n , the pair (C, A) is observable.

Observable or Detectable $\Rightarrow$ The state of the control system can be estimated using a state observer.

Coordinate Transformation Method

$$\dot{x} = Ax + Bu$$

$$y = Cx$$

$$x = T\tilde{x}$$

$$\dot{x} = T\dot{\tilde{x}}$$

(T a regular matrix)

$$\Rightarrow T\dot{\tilde{x}} = AT\tilde{x} + Bu$$

$$y = CT\tilde{x}$$

$$\Leftrightarrow \dot{\tilde{x}} = T^{-1}AT\tilde{x} + T^{-1}Bu$$

$$y = CT\tilde{x}$$

Characteristics of coordinate Transformation:

The following are preserved before and after an equivalent transformation.

- ① Transfer function
- ② Poles
- ③ Controllability and Observability

Merit of coordinate Transformation:

You can appropriately choose and consider a frame work that offers clear insights for design and analysis.